0134. Gas Station

Medium · Greedy

leetcode.com/problems/gas-station

Problem

- Circular route; gas[i] available at station i, cost[i] to reach i+1. Return the unique start index to complete the circuit, or -1.

Intuition

- If total gas < total cost, it is impossible.
- Otherwise scan once: whenever the running tank goes negative at i, no station between the current start and i can be a valid start, so move start to i + 1.

Brute Force

- Simulate from every start: O(n^2).

Optimal Approach

- Total += gas[i] - cost[i]; tank += gas[i] - cost[i].
- If tank < 0: start = i + 1, tank = 0.
- Return total < 0 ? -1 : start.

Worked Example

```
gas [1,2,3,4,5], cost [3,4,5,1,2].

i=0 diff -2 tank -2 -> start=1 tank=0
i=1 diff -2 tank -2 -> start=2
i=2 diff -2 tank -2 -> start=3
i=3 diff +3 tank 3
i=4 diff +3 tank 6
total = 0 >= 0 -> start 3
```

Java Solution

```java
public class GasStation {
    public int canCompleteCircuit(int[] gas, int[] cost) {
        int total = 0, tank = 0, start = 0;
        for (int i = 0; i < gas.length; i++) {
            int d = gas[i] - cost[i];
            total += d;
            tank += d;
            if (tank < 0) { start = i + 1; tank = 0; }
        }
        return total < 0 ? -1 : start;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- Why skipping is safe: if start..i fails, any later start in that range begins with less fuel and also fails.